Meeting 1: Continuous Time Stochastic Processes, Langevin Diffusion, and Density Evolution

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1 The Sampling Problem

In a sampling problem, we are interested in producing random values from a specified probability distribution. There are two ways we might "know" what distribution we want to sample from:

1. **Density Access** We know our distribution has un-normalized density $\pi(x) \propto e^{-U(x)}$. U is sometimes called the **energy function** from statistical physics. E.g. Normal density, "energy" is quadratic $U(x) = \frac{1}{2}\|x\|^2$, density is $\pi(x) = e^{-U(x)}$. e.x. Bayesian posterior densities, energy functions chemistry, biology, physics.
2. **Sample Access** We have a set of N example draws from the target density $\{X_i\}_{i=1}^N, X_i \sim iid\ \pi$, we use these to make more examples that look like the ones generated but are novel/ varied. e.x. language and image generators

The **score** of a density is the gradient of the log of the density, or the negative gradient of the energy function

$$s(x) = \nabla \log \pi(x) = -\nabla U(x)$$

e.x. a standard normal density has a linear score $-x$.

How can we produce samples? If we had some black-box process that simply generated samples from π, I can just use that. But we assume we cannot just "directly sample" the density we want.

Instead, we have to start by producing samples from some different density π_0 we can sample and turn them into samples from the target π.

Density Evolution

So what happens to a density when we "transform" it. Consider some initial density π_0. We consider both the density itself and a set of say N samples from that density as a histogram.

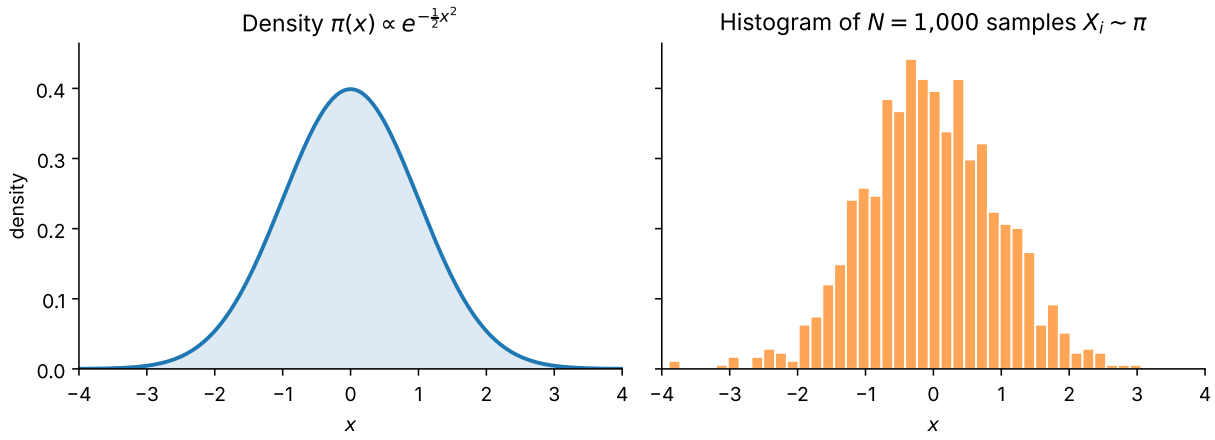


Figure 1: Two views of the same distribution $\pi_0 = N(0, 1)$: the density itself (left) and a histogram of $N = 1{,}000$ samples $X_i \sim \pi_0$ (right).

Say $X_0 \sim \pi_0$ and I apply a deterministic map to each point $X_1 = T(X_0)$. This will change the density mathematically, and also change the empirical histogram via the action on the population. E.x. $X_1 = X_0 + 3$ will shift the whole density to the right by 3 spaces. $X_1 = \sqrt{X_0}$ is a more complex change.

In general, deterministic mappings employ a known change to the density. The new density is defined by the **Jacobian Update rule**,

$$Y = T(x), \qquad p_Y(y) = p_x(T^{-1}(y))|J(T^{-1})|$$

We can also do random updates to a density. Say we define a new random variable via a conditional distribution, $p(y|x)$. Then X has marginal π_0, the new density for y is found by integrating out x

$$p_y(y) = \int p(y|x)\pi(x)dx$$

We sometimes write this update as a convolution,Y is equal to X plus noise,

$$Y = f(x) + Z, Z \sim N(0, I)$$

which means that $p(y|x) \sim N(f(x), I)$ and therefore

$$p(y) \propto \int \pi(x)e^{-\frac{1}{2}\|y-f(x)\|^2}dx$$

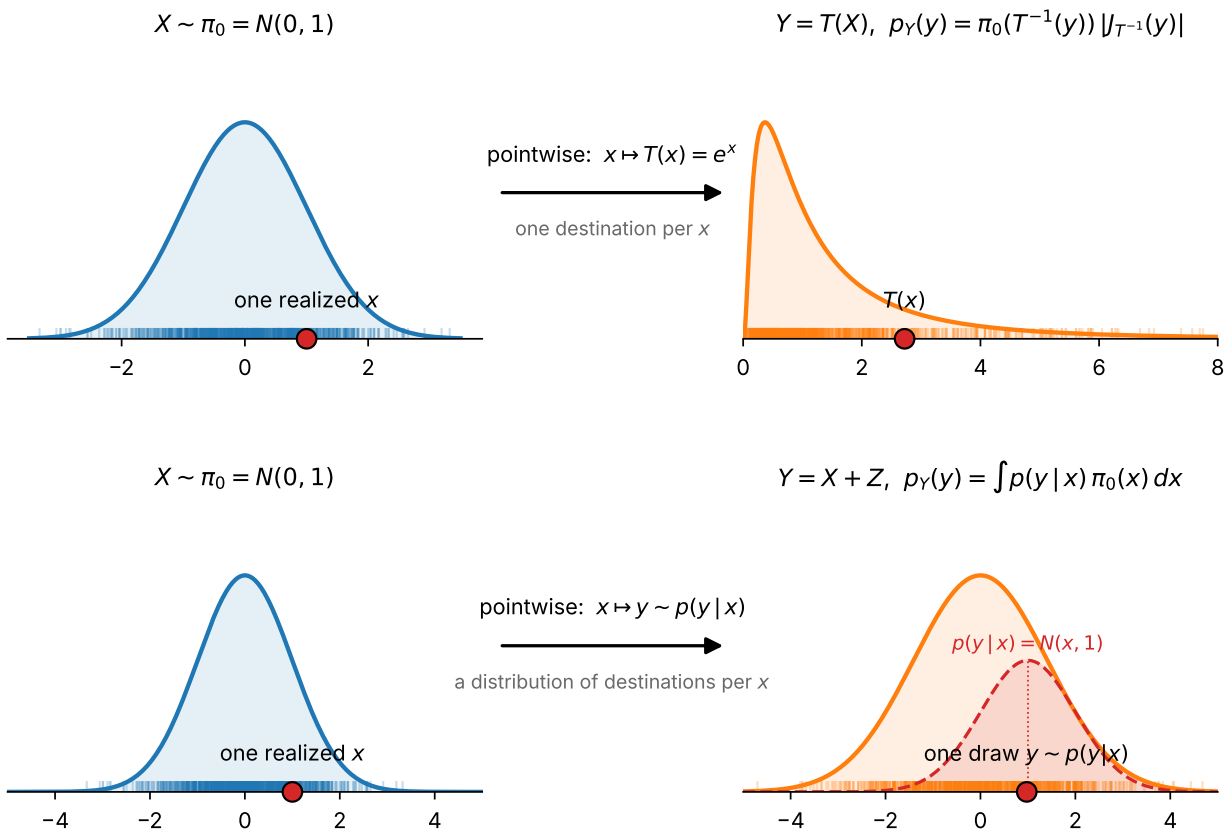


Figure 2: A deterministic map T (top) or a transition kernel $p(y|x)$ (bottom) acts on one realized x at a time, yet induces a well-defined evolution of the whole density: the Jacobian update rule and the convolution, respectively.

So a **sampling process** $X_0, X_1, \cdots, X_T$ is a sequence of random variables defined by either

- Deterministic update functions $X_t = T_t(X_{t-1})$
- Random updates via a transition kernel $P_t(X_t|X_{t-1})$

Each random variable has a marginal density $X_t \sim p_t$. We want to final density $X_T \sim p_T$ to be close to the target density. The process is designed to evolve the marginal densities to get closer to the target measure at each stage of the process.

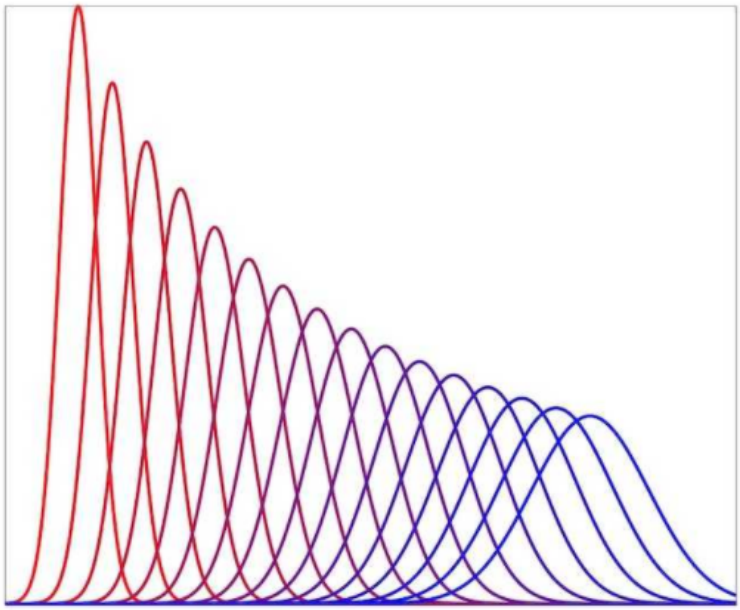


Figure 3: A sampling process as a sequence of marginal densities $p_0, p_1, \ldots, p_T$ (red to blue), each obtained from the previous one by a deterministic or random update, drifting toward the target.

This is a **Markov process** since the value of X_{t+1} only depends on the most recent value X_t and not $X_i, i < t$ (the past).

Discrepancy Functions

Discrepancy functions say how close two probability measures are to each other. Main ones: total variation, relative entropy (KL divergence), and Wasserstein-2-distance.

In general, KL divergence is the smoothest of the 3, has nice properties for analysis, and is the main one studied to say how close our sampling process is to the target density.

KL always acts as an upper bound on total variation via Pinsker inequality. A KL bound implies a TV bound.

Wasserstein is technically it's own thing, a Wasserstein bound does not imply and KL or TV and not vice versa in general. In somecases, say bounded support set or Talagrand inequality, KL is an upper bound on Wasserstein.

In general, we will focus on KL divergence and how quickly it goes to 0 relative to the target measure along the process and other discrepancy notions can be derived from that.

KL divergence is the expected log of the density ratio of two densities with respect to the first measure,

$$D_{KL}(\mu\|\nu) = \int \log(\frac{\mu(x)}{\nu(x)})\mu(x)dx$$

Another important discrepancy function is the **Fisher distance**, which is the expected difference in the scores

$$FI(\mu\|\nu) = \int \|\nabla \log \mu(x) - \nabla \log \nu(x)\|^2 \mu(x)dx$$

We will come back to the relationship between these two discrepancies later.

2 Properties of Markov Processes

We will now focus on **time invariant** update rules, which makes the same update rule at each stage of the Markov process,

$$P_t(X_t|X_{t-1}) = P(X_{t+1}|X_t) \ \forall t$$

Under fairly mild assumptions (irreducible and aperiodic), a time invariant Markov chain has a unique invariant measure. This is a density such that if $X \sim \pi^*$, and we apply the update rule, we get back the exact same density. The density is perfectly balanced to remain unchanged by the update rule, and it is the only density that has this property.

$$\pi^*(x') = \int P(x'|x)\pi^*(x)dx$$

Some facts about Markov Chains:

- They always get closer to their target density at each step of the process via data processing inequality in KL divergence

$$KL(p_{t+1}|\pi^*) \leq KL(p_t\|\pi^*)$$

- If run long enough, they will eventually converge to their invariant measure

$$\lim_{t\to\infty} P(X_t \in A) = \pi^*(A) \forall A$$

- However, we cannot run the chain to time infinity, we have to stop after say 1,000 or 1,000,000 iterations. So the final output X_T is a little bit off from the target density.

So far, we have discussed discrete time processes X_1, X_2, etc where the index is a whole number. Often, we design a sampling process first as a continuous time process $\{X_t\}_{t\geq 0}$, and then implement a discrete approximation. We do this because there are mathematical tools to define the continuous density evolution p_t as t goes from 0 to T, so we design how the density evolves in continuous time, how quickly it gets to it's invariant measure in continuous time, and then track how different the discrete process behaves from the continuous one.

3 Continuous Time Stochastic Processes Background

(see slides)

4 Langevin Diffusion

Consider a target density $\pi(x) \propto e^{-U(x)}$. Consider the SDE where the drift is the score and the dispersion is constant,

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t$$

This process is called the **Langevin Diffusion** for the target density π. I'm going to skip over the Fokker-Planck discussion for next week. But it is the case that π is the unique invariant

measure of the Langevin diffusion, and is the long run convergence measure. If run long enough, for any target π Langevin diffusion will eventually produce outputs distributed according to π.

A special case is the Langevin diffusion for a standard normal, $\pi(x) \propto e^{-\frac{1}{2}\|x\|^2}$. This process is called the **Ornstein-Uhlenbeck** process, and converges to a standard normal distribution.

$$dX_t = -X_t dt + \sqrt{2} dB_t$$

The simplest discrete approximation to the Langevin diffusion is the **Euler-Maruyama** discretization. We decide on a small step size $\delta > 0$. We then take a δ step in the direction of the drift, and add a $\sqrt{\delta}$ scaling of noise,

$$X_{t+\delta} = X_t - \delta \nabla U(X_t) + \sqrt{2\delta} Z, Z \sim N(0, I)$$

If we know the function ∇U, this is now a proper algorithm we can implement numerically. This is the **unadjusted Langevin algorithm (ULA)**. The ULA is a slightly incorrect discretization of the continuous time process, it actually does not have the exact right invariant measure, but something similar to it. The error in approximation decreases with step size, so for a small step size it is mostly sampling the correct measure.

More advanced disrecetizations of the continuous process can be considered, for example the Metropolis adjusted Langevin algorithm (MALA) or the proximal sampling algorithm, but these are not too relevant to understanding diffusion models so we won't get into these.

So the natural question to ask is

1. How quickly does the continuous process become close to it's invariant measure
2. How can we design the discrete version to become ϵ close to the target, and how many steps of the Markov chain does it take?

5 Mixing Time

Consider a Markov chain $X_1, \cdots, X_t, \cdots$ with invariant measure π^*. For any $\epsilon > 0$, the **mixing time** is the minimum number of steps of the chain required so that the marginal at time t is epsilon close to the invariant measure in TV distance,

$$t_\epsilon = \min\left(t \mid \|p_t - \pi^*\|_{TV} \leq \epsilon\right)$$

Via Pinsker inequality, KL acts as an upper bound on TV

$$\|p_t - \pi^*\|_{TV} \leq \sqrt{\frac{1}{2} KL(p_t \| p^*)}$$

as such, if $KL(p_t\|p^*) < \epsilon^2$, this implies TV distance is less that epsilon at that time point. As such, a KL bound of ϵ^2 at time T implies an upper bound on mixing time. If we can show

$$KL(p_T \| \pi^*) \leq \epsilon^2$$

this implies if I run my Markov chain for T iterations, the outputs X_T will have a TV distance from the target that is less than ϵ.

5.1 Continuous Process Mixing Time

Consider the continuous Langevin diffusion process. It is a fact that the time derivative of KL distance for Langevin diffusion is the negative Fisher distance,

$$\frac{d}{dt}KL(p_t\|\pi^*) = -FI(p_t\|\pi^*) \tag{1}$$

this is true for any target distribution π^*. Note $FI \geq 0$, so this change is always negative. So KL is always decreasing, but we already knew this via data processing inequality.

Example: Slow Mixing without LSI

This property by itself is not enough to say KL goes down quickly however. Imagine if FI gets very small and close to 0, but KL is still quite large. Then the rate of change is nearly 0, meaning KL decreases very slowly, and it takes a long time before the KL gets small. So if we want to show KL is going to 0 quickly, we need to say FI does not go to 0 too quickly.

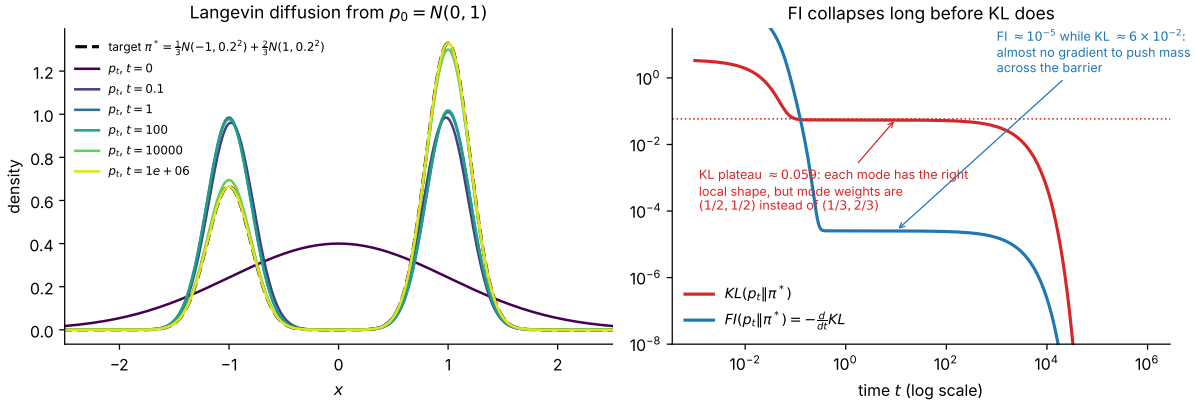


Figure 4: Langevin diffusion for the bimodal target $\pi^* = \frac{1}{3}N(-1, 0.2^2) + \frac{2}{3}N(1, 0.2^2)$ started from $p_0 = N(0, 1)$ (Fokker–Planck solved numerically). By $t \approx 1$ the mass has fallen into the two wells with the correct local shape, so the scores nearly agree and $FI(p_t\|\pi^*)$ is tiny; but the wells hold weights $(1/2, 1/2)$ instead of $(1/3, 2/3)$, so $KL(p_t\|\pi^*)$ sits on a plateau. Since $\frac{d}{dt}KL = -FI$, the KL can only drain as fast as the slow barrier crossing allows.

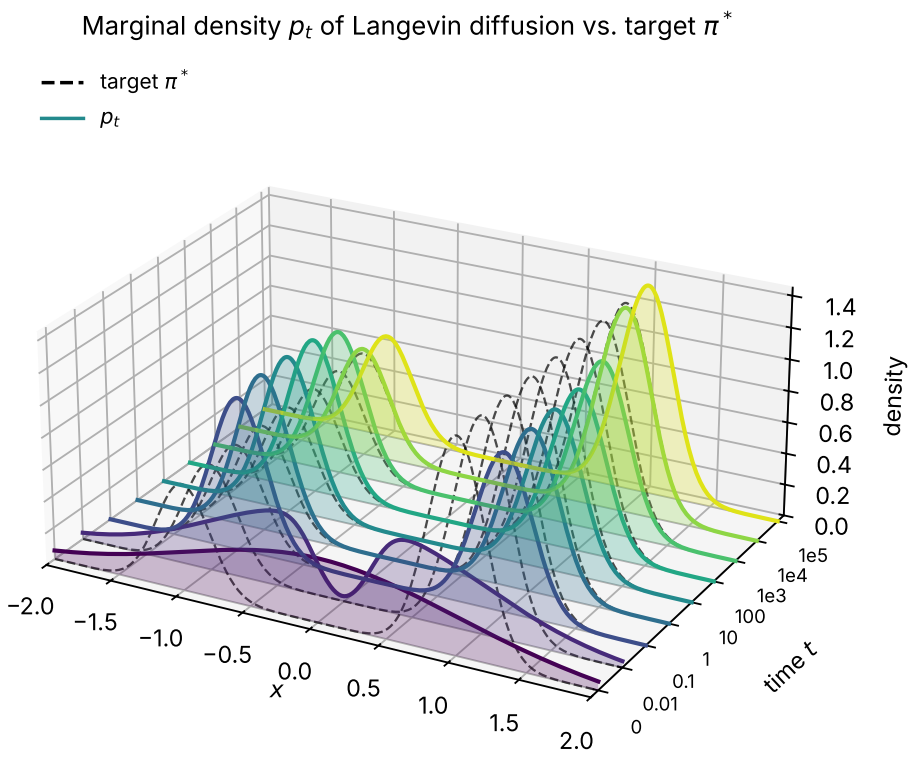


Figure 5: The same evolution with time on its own axis: the marginal p_t (solid, coloured by t) against the target π^* (dashed) at each time slice. The shape inside each well is correct almost immediately; the relative heights of the two wells take until $t \sim 10^3$–10^4 to correct.

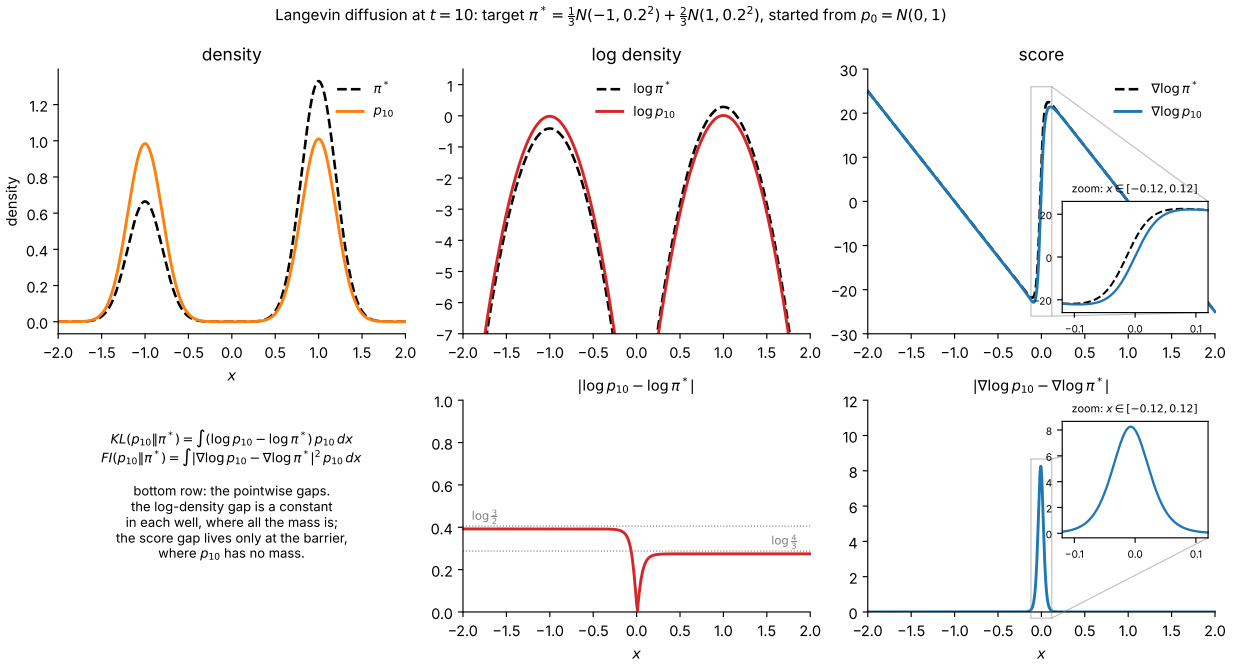


Figure 6: One snapshot on the plateau, $t = 10$. Top row: density, log density and score of p_{10} against π^*. Bottom row: the pointwise gaps. $|\log p_{10} - \log \pi^*|$ is a constant in each well ($\log \frac{3}{2}$ on the left, $\log \frac{4}{3}$ on the right), i.e. nonzero exactly where p_{10} puts its mass, so $KL(p_{10}\|\pi^*) \approx 0.06$. $|\nabla \log p_{10} - \nabla \log \pi^*|$ is zero in both wells and nonzero only in a window of width ~ 0.02 at the barrier (inset), where $p_{10} \approx 10^{-5}$, so $FI(p_{10}\|\pi^*) \approx 10^{-5}$. Fisher information is local and p_t-weighted; it cannot see that the wells have the wrong relative mass.

Rapid Mixing with LSI

A property some target densities have, but not all, is called a **Log-Sobolev inequality**. A Log-Sobolev inequality says that for a density π^* and some constant $C_{LSI} > 0$, the Fisher distance is lower bounded by the KL distance for any other distribution

$$KL(\mu\|\pi^*) \leq \frac{C_{LSI}}{2} FI(\mu\|\pi^*) \tag{2}$$

This is a special property that must be proven on a distribution-by-distribution basis. Any target density satisfies equation (1), only some densities satisfy LSI.

However, if we have an LSI then putting the two equations together, we have the expression

$$\frac{d}{dt} KL(p_t\|\pi^*) = -FI(p_t\|\pi^*) \leq -\frac{2}{C_{LSI}} KL(p_t\|\pi^*)$$

the rate of change of the KL distance is less than -1 times a multiple of itself. This implies an exponential rate of convergence

$$KL(p_t\|\pi^*) \leq e^{-\frac{2}{C_{LSI}}t} KL(p_0\|pi^*)$$

Small Log-Sobolev constant means faster convergence. If I was to run my continuous chain up to a time

$$T = \frac{C_{LSI}}{2} \log(\frac{KL(p_0 \| \pi^*)}{\epsilon^2}) = \frac{C_{LSI}}{2} \left(\log(KL(p_0 \| \pi^*)) + \log \frac{1}{\epsilon^2} \right)$$

then we would have

$$KL(p_t \| \pi^*) \leq \epsilon^2$$

So if I wanted a distance less than ϵ^2

- A larger LSI means I have to run my continuous process longer
- A larger initial discrepancy $KL(p_0 \| \pi^*)$ means I have to run the process longer
- If I want a smaller ϵ final distance, I have to run $\log(\frac{1}{\epsilon^2})$ longer, so smaller final epsilon requires longer running time

These are all properties of the continuous process. The discrete process ULA is slightly biased. As such, we have to use a small step size h and there is some additional error to control as a result of the approximation.

5.2 Discrete Process Mixing Time

The difference between discrete and continuous process errors is controlled by **Girsnov theorem**, which we will cover more in week 3 but I will skip for now.

Theorem 2.1 For the discrete process, if **Assumptions:**

1. The target density has a Log-Sobolev inequality
2. The target state space is dimension d
3. They target energy function is Lipschitz L Lipschitz,

$$\|\nabla U(x) - \nabla U(y)\| \leq L \|x - y\|$$

4. We use an appropriately small step size

$$h \leq \frac{1}{4 C_{LSI} L^2}$$

then the discrete process error decay satisfies

$$KL(p_k \| \pi^*) \leq e^{-\frac{h}{C_{LSI}} k} KL(p_0 \| pi^*) + 8 C_{LSI} L^2 dh$$

Therefore, if we run for number of steps

$$k = C_{LSI}^2 L^2 d \frac{1}{\epsilon^2} \left(\log(KL(p_0 \| \pi^*)) + \log \frac{1}{\epsilon^2} \right)$$

we will have terminal KL distance less than ϵ^2.

6 Conclusion and Next Week

- For densities that have a small Log-Sovbolev constant, a small Lipschitz constant, and we know ∇U as as function, ULA can sample quite quickly from the target density. This would be "traditional Markov Chain Monte Carlo."
- Sometimes we don't know ∇U (data access problem, image and language models), or we don't have a small C_{LSI} or Lipschitz constant (e.x. complex deep neural network posteriors)
- We run into trouble if C_{LSI} or L is exponentially large in the dimension $C_{LSI}, L \approx e^d$
- Standard normal density $e^{-\frac{1}{2}\|x\|^2}$, it turns out $C_{LSI} = 1, L = 1$. Thus OU process converges very quickly to standard normal.

The big idea of diffusion models: start from a complicated density π^*. Run the OU process **forward** which very quickly converged to standard normal, called "adding noise" to your density. Learn how to **reverse** that process, called denoising, this is the **diffusion model**. The diffusion model is a time varying Markov chain that samples the sequence of densities in reverse, starting from standard normal and finishing at the target density.

Next week we will discuss more how to construct these "reverse processes".

